

SELECT Operation in Relational Algebra and SQL

Detailed Notes with Examples

Introduction

The **SELECT** operation is fundamental in database querying. It retrieves rows (tuples) from a relation (table) based on specified criteria.

- In **Relational Algebra**, SELECT is called *Selection*, denoted by σ .
- In **SQL**, SELECT is the primary statement used to query data.

This note explains both forms with examples for better understanding.

Sample Relation: Employee

We use the following table throughout the examples:

EmpID	Name	Dept	Salary
101	Alice	HR	5000
102	Bob	IT	6000
103	Carol	IT	5500
104	Dave	Sales	4500
105	Eve	HR	5200

1. Selection in Relational Algebra (σ)

Definition

The selection operator σ extracts tuples from a relation that satisfy a specified *condition*.

$$\sigma_{condition}(R) = \{t \in R \mid t \text{ satisfies } condition\}$$

where:

- R is the input relation.
- *condition* is a predicate on attributes of R .

Example 1: Select employees from the IT department

$$\sigma_{Dept='IT'}(Employee)$$

Result

EmpID	Name	Dept	Salary
102	Bob	IT	6000
103	Carol	IT	5500

Example 2: Select employees with salary greater than 5000

$$\sigma_{Salary>5000}(Employee)$$

Result

EmpID	Name	Dept	Salary
102	Bob	IT	6000
103	Carol	IT	5500
105	Eve	HR	5200

2. Selection in SQL (WHERE clause)

Definition

In SQL, the **SELECT** statement retrieves rows from tables. The **WHERE** clause specifies conditions to filter rows, analogous to selection (σ) in relational algebra.

SELECT * FROM R WHERE condition;

Example 1: Select employees from the IT department

SQL Query

```
SELECT *  
FROM Employee  
WHERE Dept = 'IT';
```

Example 2: Select employees with salary greater than 5000

SQL Query

```
SELECT *  
FROM Employee  
WHERE Salary > 5000;
```

3. Selection with Multiple Conditions

Relational Algebra

$$\sigma_{Dept='IT' \wedge Salary > 5500}(Employee)$$

Result

EmpID	Name	Dept	Salary
102	Bob	IT	6000

SQL

SQL Query

```
SELECT *  
FROM Employee  
WHERE Dept = 'IT' AND Salary > 5500;
```

4. Selection with NOT and OR

Relational Algebra

$$\sigma_{Dept='HR' \vee Dept='Sales'}(Employee)$$

or equivalently,

$$\sigma_{NOT(Dept='IT')}(Employee)$$

Result:

Result

EmpID	Name	Dept	Salary
101	Alice	HR	5000
104	Dave	Sales	4500
105	Eve	HR	5200

SQL

SQL Query

```
SELECT *  
FROM Employee  
WHERE Dept = 'HR' OR Dept = 'Sales';
```

-- or

```
SELECT *  
FROM Employee  
WHERE NOT Dept = 'IT';
```

5. Selection on NULL and Other Conditions

SQL

- To select rows where an attribute is NULL, use IS NULL or IS NOT NULL.

SQL Query

```
SELECT *  
FROM Employee  
WHERE ManagerID IS NULL;
```

Note: Relational algebra does not directly model NULL values; this is a practical SQL extension.

6. Projection combined with Selection

Often, selection is combined with projection to select specific columns of the filtered rows.

Relational Algebra

Example: Names of employees in IT department

$$\pi_{Name} (\sigma_{Dept='IT'}(Employee))$$

Result:

Result	
Name	
Bob	
Carol	

SQL

SQL Query
SELECT Name FROM Employee WHERE Dept = 'IT';

Summary

- **Selection** in relational algebra is expressed by the operator σ with a predicate condition.
- **SQL SELECT** combined with **WHERE** clause performs the same filtering operation.
- Multiple conditions can be combined using logical operators **AND**, **OR**, and **NOT** (or $\wedge$, $\vee$, $\neg$ in relational algebra).
- Selection is often combined with projection (π) to retrieve specific columns.
- SQL supports **NULL** handling which is not represented in pure relational algebra.